A NEW THEORY OF CREDIT LINES (WITH EVIDENCE)

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FACTS

Credit lines make up c. \$2T of committed credit, bulk of bank credit (80%)
Berg–Saunders–Steffen 20, Greenwald–Krainer–Paul 21, Chodorow-Reich et al. 21, and Sufi 09

Credit line utilization low (zero for 45% of firms)
Ivashina–Scharfstein 10, Greenwald 21

Credit lines are bundled with loans, especially to risky firms ($> 70\%$)
See below

QUESTIONS

Q1. Why are credit lines so common, even if rarely used?

Q2. Why are credit lines bundled with loans?

THIS PAPER

Dynamic model of borrower B issuing debt
Admati et al 17/DeMarzo–He 21

Friction: Non-exclusivity

After borrowing from one lender at t, borrows from another at $t + \mathrm{dt}$

Innovation: Allow for credit lines (CLs)

Options to borrow $\tilde{p}\mathrm{d}\tilde{Q}$ against face value $\mathrm{d}\tilde{Q}$

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P7. Deterrence (via two experiments): $\uparrow$ CL availability/credibility $\downarrow$ debt

MODEL

OVERVIEW

Infinite-horizon sequential borrowing: B borrows from one lender at each t

Credit line-debt bundle at date 0 and new debt afterward

B's cost of debt $c(Q_t)$ increasing and concave in stock of debt Q_t

Captures expected coupon payment $\downarrow$ as default prob. $\uparrow$ in Q

Universal risk-neutrality, deep pockets, and discounting at rate ρ

NB: No revocation; no risk; no CLs at $t > 0$

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B's flow payoff: $v_t \mathrm{d}t = y\mathrm{dt} + p_t \mathrm{d}Q_t - c(Q_t)\mathrm{d}t$, where

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B's value: $V_t = \int_0^\infty e^{-\rho s} v_{t+s} \mathrm{d}s$

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Lenders' flow payoff from unit debt given stock Q_t: expected coupon $\gamma(Q_t)$

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Value of stream of coupons on unit of debt: $\Gamma_t = \int_0^\infty e^{-\rho s}\gamma(Q_{t+s})\mathrm{d}s$

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($\mathrm{d}\tilde{Q}$ put options on debt with strike $\tilde{p}$)

ASSUMPTIONS

A1: Dilution: expected coupons lower for higher debt: $\gamma' < 0$ & $\gamma(\infty) = 0$

A2: Gains from trade:

Positive at $Q = 0$: $\gamma(0) > c'(0)$

Negative as $Q \to \infty$: $\gamma(\infty) < c'(\infty)$

A3: Curvature: c and $\gamma(Q)Q - c(Q)$ concave

SOLUTION CONCEPT: MARKOV PERFECT EQ.

At each date, lenders offer bundles/loans and B chooses one offer s.t.

B and lenders max expected future lifetime utility given beliefs

Beliefs are consistent

B's balance sheet $\left(Q_t, (\tilde{p}, \mathrm{d}\tilde{Q})\right)$ is suff. stat. for history w.r.t. action

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Notation: $\tilde{V}$ value function with credit line; V without (after draw down)

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$$c(Q) = \underbrace{(1-\pi(Q))\delta Q}_{\mathbb{E}[\text{coupons}]=\gamma(Q)Q} - \underbrace{\tau_c \delta Q}_{\text{tax shield}} + \underbrace{\pi(Q)Qk}_{\mathbb{E}[\text{default costs}]}$$

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Baseline assumptions easy to satisfy

BENCHMARKS

BENCHMARK 1: EXCLUSIVE COMPETITION

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If B commits to one lender (of those competing at date 0) forever

B borrows at Date 0 and never again

Price is above marginal cost

EXCLUSIVE COMP: PROGRAM W/ COMMITMENT

Choose $(Q_t)_{t\geq 0}$ at $t = 0$ to max V_0 s.t. $p_t \leq \Gamma_t$

EXCLUSIVE COMP: PROOF

Part 1: Optimal allocation $\implies$ no borrowing for $t > 0$

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Max $p_0 Q_0 - \dfrac{c(Q_0)}{\rho}$ s.t. $p_0 \leq \dfrac{\gamma(Q_0)}{\rho}$

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Cf. Leverage ratchet effect and Coase Conjecture
(Admati et al. 18, DeMarzo–He 21, Coase 72,...)

NO CLs: PROGRAM (W/O COMMITMENT)

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NB: c concave $\implies$ V convex $(V'' = -c''/\rho > 0)$

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After selling monopoly Q to one lender, profit by selling more to another

RESULTS

LEMMA: RATCHET EFFECT FOR CREDIT LINES

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Suppose B has CL $(\tilde{p}, \mathrm{d}\tilde{Q})$ in place s.t. indifferent to drawing at Q_0:

$$\tilde{p}\mathrm{d}\tilde{Q} + V(Q_0 + \mathrm{d}\tilde{Q}) = \tilde{V}(Q_0) \quad \Longleftrightarrow \quad \tilde{p} = -\frac{V(Q_0 + \mathrm{d}\tilde{Q}) - \tilde{V}(Q_0)}{\mathrm{d}\tilde{Q}}$$

B weakly prefers to draw for $Q > Q_0$

RATCHET: PROOF SKETCH FOR SMOOTH Q_t & $\tilde{V}$

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Sub into condition not to draw in terms of V:

$$
\begin{aligned}
\tilde{p} &< -\frac{V(Q_0^+ + \mathrm{d}\tilde{Q}) - \tilde{V}(Q_0^+)}{\mathrm{d}\tilde{Q}} && \text{(preference not to draw at } Q_0^+\text{)} \\
&= -\frac{V(Q_0^+ + \mathrm{d}\tilde{Q}) - V(Q_0^+)}{\mathrm{d}\tilde{Q}} && (\tilde{V}(Q_0^+) = V(Q_0^+) \text{ from above}) \\
&< -\frac{V(Q_0 + \mathrm{d}\tilde{Q}) - V(Q_0)}{\mathrm{d}\tilde{Q}} && \text{(convexity of } V \text{ from BM2)} \\
&= \tilde{p} && \text{(indifference at } Q_0\text{)}
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I.e. $\tilde{p} < \tilde{p}$: $\Rightarrow\Leftarrow$

RATCHET EFFECT OF CLs: INTUITION

c concave $\implies$ more debt B has, less costly to have more

Idea: Higher $Q \implies$ lower repayment prob. $\implies$ lower cost of $\mathrm{d}\tilde{Q}$

Akin to leverage ratchet effect:

Higher debt begets higher debt

R1: RATCHET-ANTI-RATCHET EFFECT

R1: RATCHET-ANTI-RATCHET EFFECT

Suppose B has debt Q_0 and CL $(\tilde{p}, \mathrm{d}\tilde{Q})$ in place s.t. indifferent to drawing

If $\mathrm{d}\tilde{Q}$ large enough, B doesn't take any new debt ($q_t \equiv 0$)

R1: RATCHET-ANTI-RATCHET EFFECT: PROOF

Step 1: Find price s.t. B prefers to borrow dQ at p (and thus draw CL)

Step 2: Find price s.t. lender willing to lend dQ at p

Step 3: Show price B willing to borrow $>$ price lenders willing to lend

R1: STEP 1: FIND p S.T. B PREFERS TO BORROW

Issuing dQ at p (and drawing CL) $\succ$ not issuing (and $\sim$ drawing) if

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Or, for $\mathrm{d}t \to 0$, $p > -\dfrac{V(Q_0 + \mathrm{d}Q + \mathrm{d}\tilde{Q}) - V(Q_0 + \mathrm{d}\tilde{Q})}{\mathrm{d}Q}$

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Must hold for $\mathrm{d}Q \to 0 \implies p > -V'(Q_0 + \mathrm{d}\tilde{Q}) = \dfrac{c'(Q_0 + \mathrm{d}\tilde{Q})}{\rho}$

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Must hold for $\mathrm{d}Q \to 0 \implies p > -V'(Q_0 + \mathrm{d}\tilde{Q}) = \dfrac{c'(Q_0 + \mathrm{d}\tilde{Q})}{\rho} > \dfrac{\bar{c}'}{\rho} > 0$

R1: STEP 2: FIND p S.T. LENDER LENDS

Lender knows B draws if lends, so lends $\mathrm{d}Q$ at p iff $p \leq \Gamma(Q_0 + \mathrm{d}Q + \mathrm{d}\tilde{Q})$

$\Gamma(Q_0 + \mathrm{d}Q + \mathrm{d}\tilde{Q}) \to 0$ as $\mathrm{d}\tilde{Q} \to \infty$ since $\gamma(Q) \to 0$ as $Q \to \infty$ by A2

R1: STEP 3 NO LENDING

Step 1: B borrows if $p > \dfrac{c'(Q_0 + \mathrm{d}\tilde{Q})}{\rho} > \dfrac{\bar{c}'}{\rho}$

Step 2: Lender lends if $p \leq \Gamma(Q_0 + \mathrm{d}\tilde{Q}) \to 0$

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R1: RATCHET-ANTI-RATCHET: INTUITION

When B borrows from new lenders, draws on credit line diluting them

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NB: Self-enforcing independent of who holds debt

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NB: Self-enforcing independent of who holds debt

Debt can be traded, as in practice, unlike in monitoring-based theories

R1 RESONATES WITH PRACTICE

Credit line latent off-equilibrium threat

Explains low utilization: zero for 45% of firms; 16% for median

Credit line large so price falls enough after drawn

Explains large size of undrawn CLs: 29% of liabilities, 40% of equity

R2: BUNDLING

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At date 0, B chooses a bundle of a loan (p, Q_0) and CL $(\tilde{p}, \mathrm{d}\tilde{Q})$ s.t.

(p, Q_0) coincides with outcome of exclusive competition

$(\tilde{p}, \mathrm{d}\tilde{Q})$ makes B indifferent to drawing at Q_0 with $\mathrm{d}\tilde{Q}$ "large"

R2: BUNDLING: PROOF

Recall that under candidate bundle

B never borrows again by R1

B's payoff maximized s.t. lenders break even by BM1

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B never borrows again by R1

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$\Longrightarrow$ At any other bundle another lender can profitably enter

Can post candidate with price p replaced by $p + \epsilon$

R2: BUNDLING: INTUITION

Credit lines allow B to commit not to dilute by R1

Use credit lines to implement optimum without dilution (i.e. BM1)

R2 RESONATES WITH PRACTICE

Credit line bundled with loan as commitment device to curb dilution

Explains bundling, esp. for risky firms with high dilution risk

CLs OFFERED IN DEBT BUNDLES

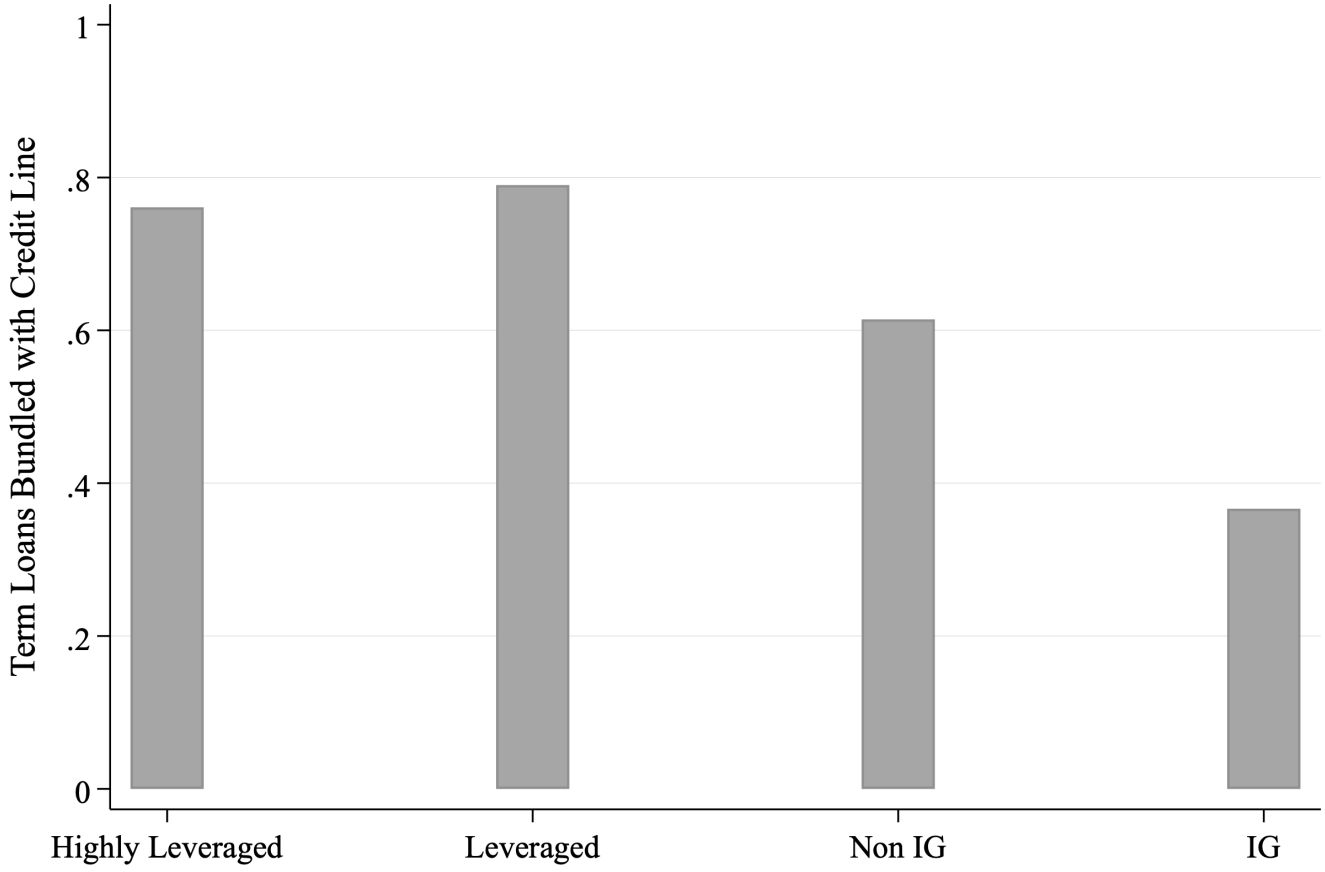


Figure: Bundling Propensity and Firm Riskiness

This figure shows how the proportion of term loans that are bundled with a credit line vary by firm riskiness, as measured by Dealscan's classification of firms' market segments. The loans to the safest borrowers are Investment Grade ("IG"), then "Non-IG", "Leveraged", and "Highly Leveraged" respectively, where the distinction between the latter three categories depends on pricing thresholds. Data are from Dealscan, covering US C&I syndicated loans from 1982 through 2021 and excluding financials.

EXT. 1: REVOCATION RISK

EXTENSION 1: LENDER REVOCATION RISK

So far, lenders always honor credit lines

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So far, lenders always honor credit lines

Now lender honors w.p. α and revokes it w.p. $1 - \alpha$

EXT 1: REVOCATION RISK AND BORROWING

Higher revocation risk $1 - \alpha \implies$ higher debt

EXT 1: REVOCATION RISK: INTUITION

If CL revoked, new lender not diluted by draw

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EXT. 2: COMPETITION

EXTENSION 2: LENDER COMPETITION

Suppose fixed cost κ of bundling

EXTENSION 2: LENDER COMPETITION

Suppose fixed cost κ of bundling

After contracting, B non-exclusive w.p. $1 - \mu$, exclusive w.p. μ

EXT 2: COMPETITION

More competition $\implies$ more bundling

EXT 2: COMPETITION: INTUITION

Value of bundling higher if prob. of non-exclusivity $1 - \mu$ high

EXT. 3: DEFAULT RISK

EXTENSION 3: BORROWER DEFAULT RISK

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Suppose fixed cost κ of bundling

EXTENSION 3: BORROWER DEFAULT RISK

Suppose fixed cost κ of bundling

Vary sensitivity of default to new debt: $\left(\text{e.g. } -\gamma'(Q_0) = \delta\pi'(Q_0)\right)$

EXT 3: DEFAULT RISK AND BUNDLING

Higher $-\gamma'(Q_0) \implies$ higher bundling

EXT 3: DEFAULT RISK: INTUITION

$$\frac{\gamma'(Q_0)Q_0}{\rho} = p_0 - \text{MC: Monopolist price wedge}$$

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Higher wedge $\implies$ higher impact of new debt $\implies$ higher dilution risk

$\implies$ more value of exclusivity $\implies$ more willing to pay κ to bundle

PREDICTIONS & TESTS

PREDICTION 1: BUNDLING

Prediction: Credit lines are bundled with loans

Reason: CL protects loan from dilution

Evidence: $> 70\%$ term loans bundled with CL at origination

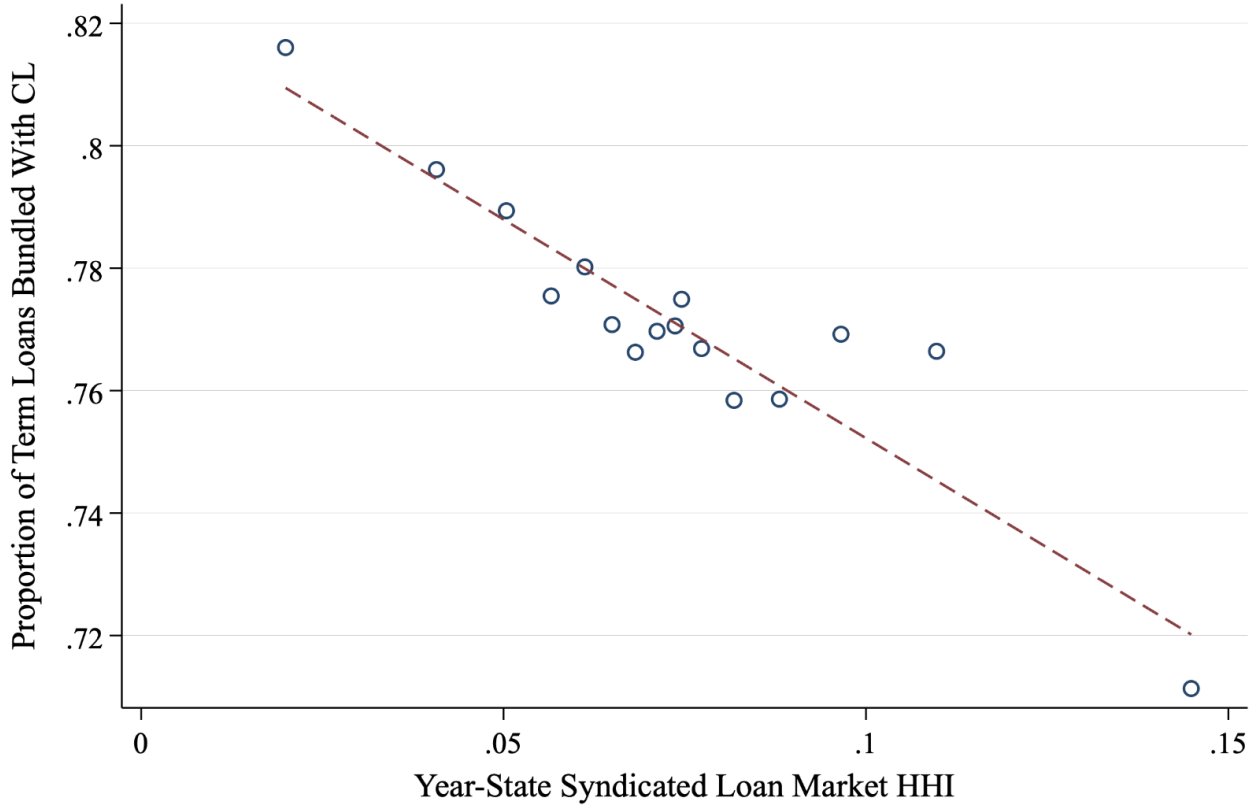
PREDICTION 2: BUNDLING PROPENSITY

Prediction: Bundling propensity higher if dilution risk higher

Reason: Protection valuable only if dilution likely

Evidence: More bundling if default likely or lenders less concentrated

PR2: BUNDLING ↑ AS CONCENTRATION ↓



This figure plots a bin scatter of the following regression for lender l, bundling propensity$_{lst} = \alpha_{lt} + \beta \text{HHI}_{st} + \varepsilon_{lst}$, where bundling propensity$_{lst}$ is the proportion of term loan deals that include a credit line among deals that lender l originates in state s in year t; HHI$_{st}$ is the syndicated loan market HHI in state s in year t; and α_{lt} is a lender-year fixed effect. Data are from Dealscan, covering US C&I syndicated loans from 2002 through 2019 and excluding financials.

PREDICTION 3: SIZE

Prediction: Credit lines are large

Reason: CL deters new borrowing only if large

Evidence: Undrawn CLs are

80% of book equity, 20% of market equity, 1.7 times the size of loan

PREDICTION 4: SIZE IN CROSS-SECTION

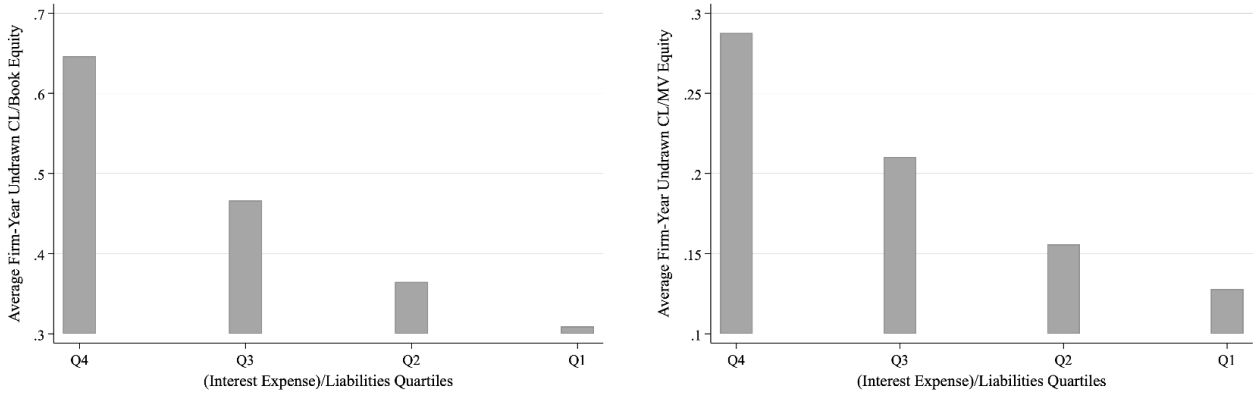
Prediction: Credit lines are larger if dilution risk higher

Reason: Deterrant more valuable when dilution more likely

Evidence: Larger CLs if default likely or lenders less concentrated

PR3: CL SIZE ↑ AS DEFAULT ↑

Figure 3: Credit Line Size is Increasing in Firm Riskiness

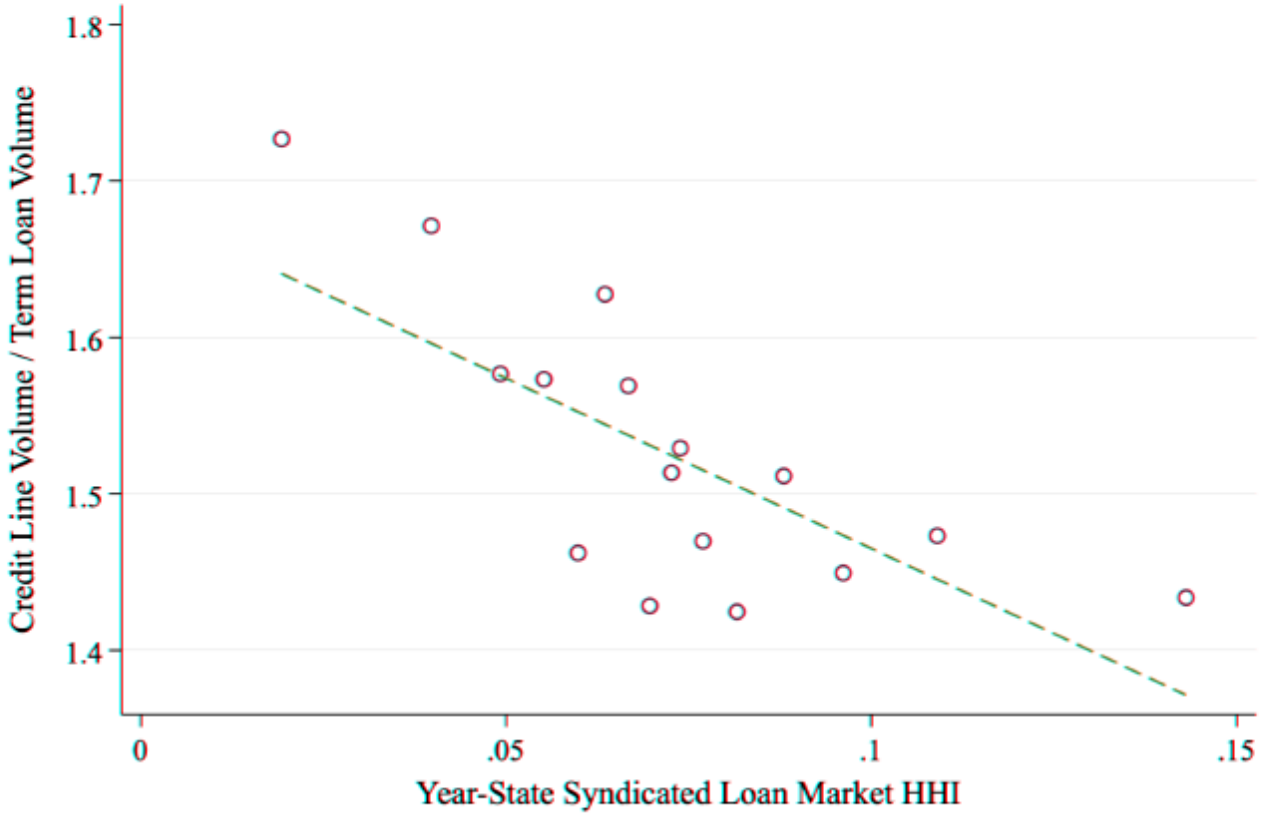


(a) As Share of Book Equity (b) As Share of Market Equity

This figure plots the average firm-year undrawn CL as a share of (a) book equity and (b) market equity, by quartiles of (firm interest expense)/(total liabilities), and where Q4 represents firms with the highest interest expense. Data are annual at the firm level from Capital IQ and Compustat from 2002 through 2019, covering US public non-financial firms. Stock price data comes from CRSP.

PR3: CL SIZE ↑ AS CONCENTRATION ↓

Figure 4: Credit Line Size is Increasing in Competition



This figure plots a bin scatter of the following regression specification for lender l, $\left(\frac{\text{credit line volume}}{\text{term loan volume}}\right)_{lst} = \alpha_{lt} + \beta \text{HHI}_{st} + \varepsilon_{lst}$, where $(\text{credit line volume/term loan volume})_{lst}$ is the average volume of lender l's credit lines relative to the bundled term loans in state s in year t HHI_{st} is the syndicated loan market HHI in state s in year t; and α_{lt} is a lender-year fixed effect. Data are from Dealscan from 2002 through 2019. Average credit line volume is winsorized at 2%.

PREDICTION 5: UTILIZATION

Prediction: Credit lines are rarely drawn

Reason: CL off-equilibrium threat

Evidence: 45% of borrowers utilize none of CL; median 17%

PREDICTION 6: RATES

Prediction: Rates on CLs relatively low

Reason: Rate s.t. lender worse off when lends

Evidence: Median rate on CL $\approx$ that on the associated term loan

PREDICTION 7: BORROWING

Prediction: Credit lines deter subsequent borrowing

Reason: Anti-ratchet

Evidence: Increased prob. CL revoked increases borrowing

TEST OF PREDICTION 7

TEST

Based on fact that unhealthy lenders more likely to revoke credit lines (Chodorow-Reich–Falato 22)

Construct (neg) health shocks for borrowers' CL lenders and all lenders (following Chodorow-Reich 14 and Darmouni 20)

Shocks uncorrelated with borrower characteristics

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$\gamma < 0$: Shocks to all lenders decrease it (in line with literature)

$\beta > 0$: Shocks to CLs increase borrowing (in line with our theory)

INCREASE DEBT AFTER SHOCKS TO CLs

	(1)	(2)	(3)	(4)
Shock	-0.16^{***}	-0.17^{***}	-0.17^{***}	-0.18^{***}
	(0.05)	(0.05)	(0.05)	(0.05)
Shock CL		0.03^{***}	0.02^{**}	0.03^{***}
		(0.01)	(0.01)	(0.01)
Number of Syndicates			0.03^{***}	0.04^{***}
			(0.01)	(0.01)
Pre CL Indic				-0.03^{**}
				(0.01)
Constant	0.20^{***}	0.19^{***}	0.19^{***}	0.18^{***}
	(0.04)	(0.04)	(0.04)	(0.04)
Observations	4883	4883	4883	4883
Adjusted R^2	0.002	0.003	0.009	0.010

ROBUSTNESS

CREDIT LINE BUNDLES FOR $t > 0$

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So far: $t > 0$ lenders offer only debt, no new credit lines

Now: New lender can offer loan-CL bundle

R3: NO NEW CL LENDERS

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Suppose equilibrium with loan-credit line pair outstanding

No new lender offers bundle that B would accept

R3: NO NEW CL LENDERS: PROOF (SKETCH)

Same argument: Lender knows B will draw line in place (ratchet effect)

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NB: Only sketch; full proof shows ratchet effect robust to multiple CLs

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NB: Only sketch; full proof shows ratchet effect robust to multiple CLs

Show by bounding payoff by commitment payoff

R4: WITH RISK, CONTINGENT CLs IMPLEMENT OPTIMUM

ADDING RISK

So far: No risk

Here: Allow for risk

How: Allow for additional Markov state s with debt Q_s contingent on s

Let Q_s^* be optimal debt at s

Let V_s and $\tilde{V}_s$ be value functions with and w/o CL

R4: CONTINGENT CLs IMPLEMENT OPTIMUM

Suppose V and $\tilde{V}$ exist, are convex, and monotone

Let $\tilde{p}_s = -\dfrac{V_s(Q_s^* + \mathrm{d}\tilde{Q}) - \tilde{V}_s(Q_s^*)}{\mathrm{d}\tilde{Q}}$ with $\mathrm{d}\tilde{Q}$ large

A contract with one lender with Q_s^* and $(\tilde{p}_s, \mathrm{d}\tilde{Q})$ implements optimum

R4: PROOF (SKETCH)

By definition, B indifferent to drawing at Q_s^* in state s

By ratchet, B draws if $Q > Q_s^*$

By $\mathrm{d}\tilde{Q}$ large, other lenders don't lend ($\Gamma(\mathrm{d}\tilde{Q}) \to 0$)

CONTRIBUTIONS

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Credit lines:

In lit.: Insurance to take on more debt

Here: Commitment to take on less debt

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In lit.: Excessive debt and zero surplus

Here: Allowing for credit lines makes ratchet effect self-deterring

Latent contracts:

In lit.: Help lenders support collusive outcomes

Here: Help borrowers support monopolistic outcome

CONCLUSION

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Heretofore unexplored role of credit lines: commit not to dilute

Curb competition with future self

Implement monopoly outcome

A NEW THEORY OF CREDIT LINES

$$q > 0$$

$q > 0$

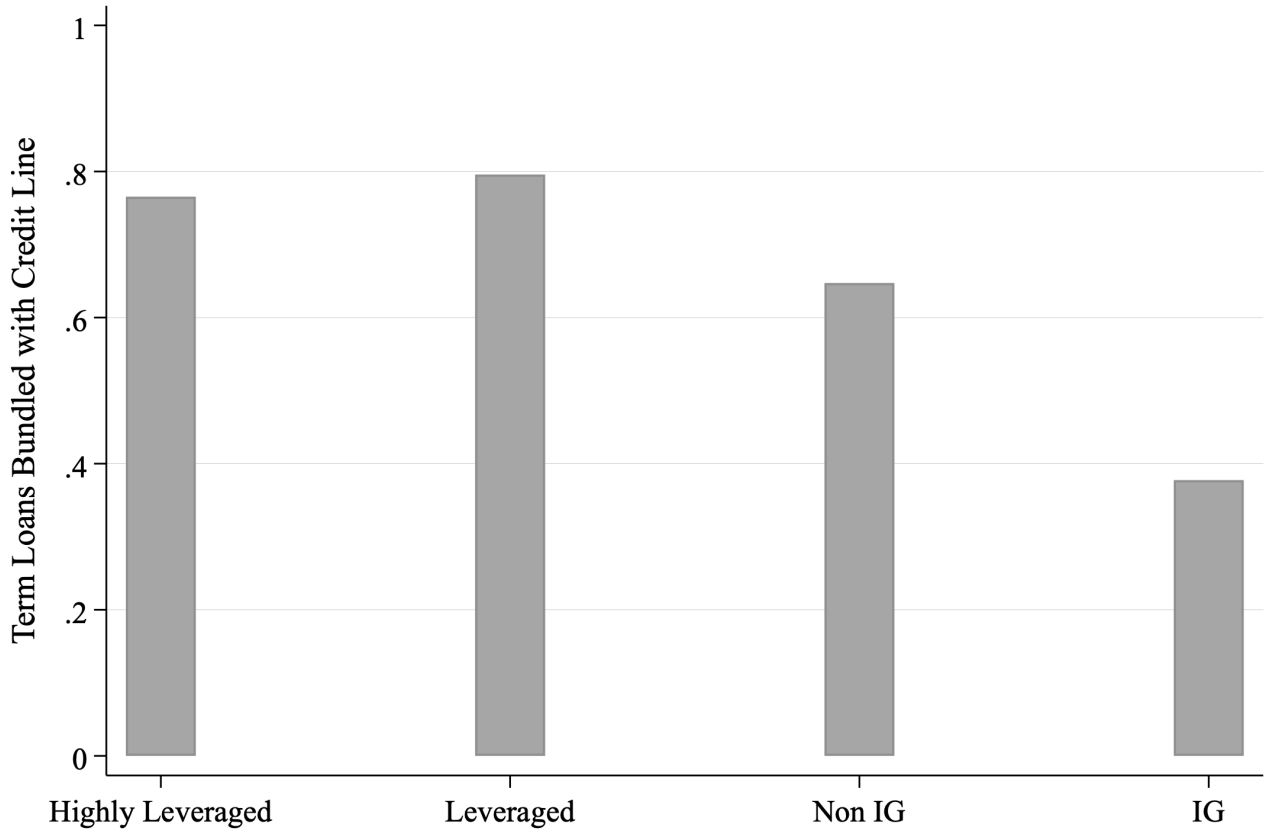
p satisfies B–S "PDE": $\rho p(Q) = \gamma(Q) + p'(Q)q$ where $p(Q) = \dfrac{c'(Q)}{\rho}$

$$\implies q = \frac{\gamma(Q) - c'(Q)}{-c''(Q)/\rho} > 0 \text{ by A2 and A3}$$

(Go back)

BUNDLING AND FIRM RISK

Figure: BUNDLING AND FIRM RISK



Data from Dealscan: US C&I syndicated loans from 1997–2021 to non-financial firms with ≥ 1 US lender a US bank

(Go back)

DATA

DEALSCAN SYNDICATED LOAN DATA

Restrict to deals originated in US, exclude loans to financials

Keep C&I loans: deal purpose is general purpose or working capital

Impute lenders' syndicate weights following Chodorow-Reich 14

Controls

- Number of syndicates firm i borrowed from during normal period
- Indicator variable tracking if firm borrowed CL in normal period

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SHOCK CONSTRUCTION

CONSTRUCTION OF SHOCKS

Dealscan US C&I syndicated loan data, following Chodorow-Reich 14 and Darmouni 20

$\Delta L_{b,-i}$: decrease b's lending to $j \neq i$ in crisis $\Delta L_{b,-i} := 1 - \dfrac{2\sum_{j\neq i} \alpha_{b,j,\text{cris}} L_{b,j,\text{cris}}}{\sum_{j\neq i} \alpha_{b,j,\text{norm}} L_{b,j,\text{norm}}}$

Normal: 1/2004–8/2008; crisis: 10/2008–12/2010

$L_{b,j}$: # loans from b to j

$\alpha_{b,j}$ average syndicate weight

i's shock: sum $\Delta L_{b,-i}$ over lenders last pre-crisis syndicate S, weighted by α_s:

$$\text{Shock CL}_i = \sum_{b\in S_{CL}} \alpha_s^{CL} \Delta L_{b,-i} \qquad\qquad \text{Shock}_i = \sum_{b\in S} \alpha_s \Delta L_{b,-i}$$

Shock $\text{CL}_i = 0$ if firm i's last pre-crisis syndicate has no CL

S_{CL} is a syndicate that originates a deal including a CL

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